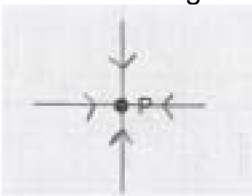
1(a)	at least 4 straight radial lines to P 	**B1**
	all arrows pointing along the lines towards P	**B1**
1(b)	**Any 2 from:** gravitational force provides the centripetal force (centripetal or gravitational) force has constant magnitude (centripetal or gravitational) force is perpendicular to velocity (of moon) / direction of motion (of moon)	**B2**
1(c)(i)	$\frac{GMm}{r^2} = mr\omega^2$	**M1**
	$M = \frac{r^3\omega^2}{G}$ and gradient = $r^3\omega^2$ hence $M = \frac{\text{gradient}}{G}$ or $r^3 = GM \times 1/\omega^2$ so gradient = GM hence $M = \frac{\text{gradient}}{G}$	**A1**
1(c)(ii)	M = $4.1 \times 10^{23} / (6.0 \times 10^{7} \times 6.67 \times 10^{-11})$ = $\mathbf{1.0 \times 10^{26}}$ kg	**B1**

Question	Answer	Marks
1(c)(iii)	$\frac{GMm}{r^2}=\frac{mv^2}{r}$ $\frac{GM}{r}=v^2$	**C1**
	$v^2=\frac{6.67\times10^{-11}\times1.0\times10^{26}}{1.2\times10^{8}}$ $v^2=5.6\times10^{7}\text{m s}^{-1}$	**C1**
	$v=7500\text{ m s}^{-1}$	**A1**

Question	Answer	Marks